

[5254] - 550

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(Elective - IV) (2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Draw suitable neat diagrams, wherever necessary.
- 2) Figures to the right indicate full marks.
- 3) Use of electronic pocket calculator is allowed.
- 4) Assume suitable data, if required.

Q1) a) Write short notes on : [6]

- i) Effect of element aspect ratio on accuracy.
- ii) Mesh refinement vs higher order elements.

b) Discuss the advantages and disadvantages of FEM over [4]

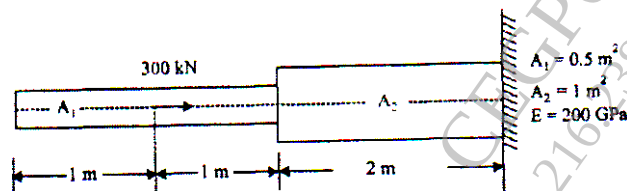
- i) Classical method
- ii) Finite difference method

OR

Q2) a) Explain the principle of Rayleigh - Ritz method. [6]

b) Explain plane stress formulation and its importance. [4]

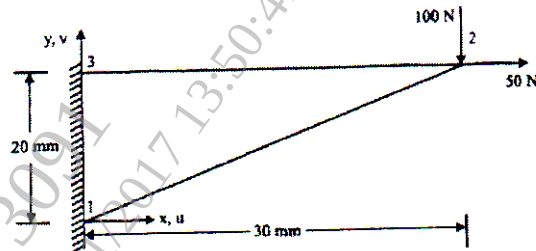
Q3) Determine the nodal displacements and element stresses by finite element formulation for the following figure. Use $P = 300 \text{ kN}$; $A_1 = 0.5 \text{ m}^2$; $A_2 = 1 \text{ m}^2$; $E = 200 \text{ GPa}$ [10]



OR

P.T.O.

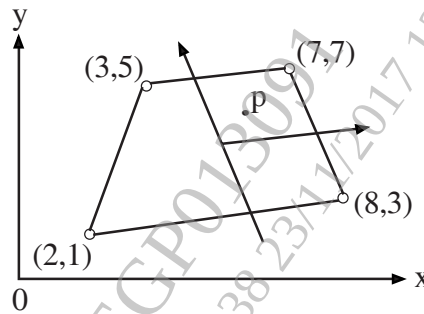
- Q4) a)** Calculate displacements and stress in a triangular plate, fixed along one edge and subjected to concentrated load at its free end. Assume $E = 70,000 \text{ MPa}$, $t = 10 \text{ mm}$ and $\nu = 0.3$ [6]



- b) What is meant by displacement function? Write down convergence criteria for Finite element Analysis. [4]

- Q5) a)** Explain the isoparametric concept in finite element analysis. [6]

- b) Determine the Cartesian coordinate of the point $P(\zeta = 0.25, \eta = 0.5)$ shown in fig. [6]

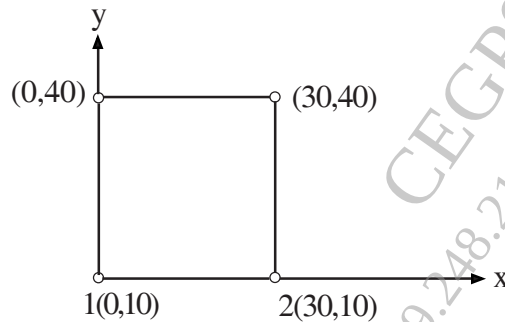


- c) State and explain the three basic laws on which isoparametric concept is developed. [6]

OR

- Q6) a)** Explain the terms isoparametric, subparametric and superparametric elements. [8]

- b) For the element shown in fig. assemble Jacobian matrix and strain displacement matrix for the Gaussian point $(0.2578, 0.6550)$. [10]

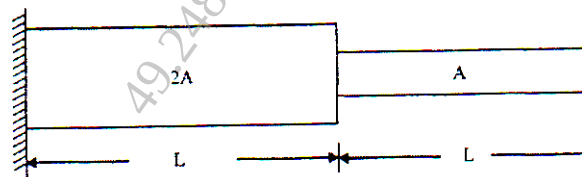


- Q7)** a) Write down governing equation of steady state Heat Transfer and also write down elemental stiffness matrix and compare with Bar element. [6]
- b) Consider a brick wall of thickness 0.6 m, $k = 0.8 \text{ W/m}^\circ\text{K}$. The inner surface is at 28°C and the outer surface is exposed to cold air at -10°C . The heat transfer coefficient associated with the outside surface is $40 \text{ W/m}^2\text{K}$. Determine the steady state temperature distribution within the wall and also the heat flux through the wall. Use two elements and obtain the solution. [10]

OR

- Q8)** a) Heat is generated in a large plate ($K = 0.4 \text{ W/m}^\circ\text{C}$) at the rate of 5000 W/m^3 . The plate is 20 cm thick. Outside surface of the plate is exposed to ambient air at 30°C with a convective heat transfer coefficient of $20 \text{ W/m}^2\text{C}$. Determine the temperature distribution in the wall. [10]
- b) Derive FEA stiffness matrix for Pin Fin Heat Transfer problem. [6]

- Q9)** a) Write down Consistent mass and Lumped Mass Matrix for [6]
- i) Bar Element
- ii) Plane Stress Element
- b) Find the natural frequencies of longitudinal vibrations of the same stepped shaft of areas $A = 1000 \text{ mm}^2$ and $2A = 2000 \text{ mm}^2$ and of equal lengths ($L = 1\text{m}$), when it is constrained at one end, as shown below. [10]



OR

- Q10)** a) Explain Each term of Dynamic Equation given below and explain importance of eigen values and eigen vectors. [6]

$$m\ddot{x} + c\dot{x} + kx = f$$

- b) Find the natural frequencies of longitudinal vibrations of the unconstrained stepped shaft of areas A and $2A$ and of equal lengths (L), as shown below. [10]

